

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS HSC ASSESSMENT TASK 3 JUNE 2009

Time Allowed: 70 minutes

Instructions:

- Write using blue or black pen.
- Approved calculators may be used.
- Attempt all questions.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new side of a page.
- A table of standard integrals is supplied.

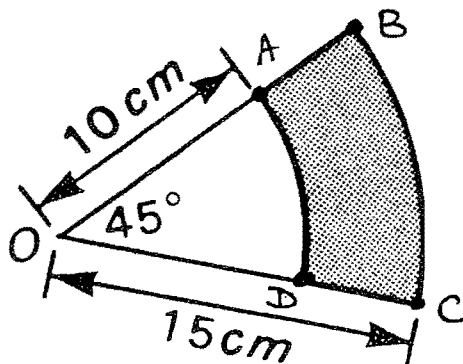
Name:

Q1	Q2	Q3	Q4	Q5	Total
/12	/12	/12	/12	/12	/60

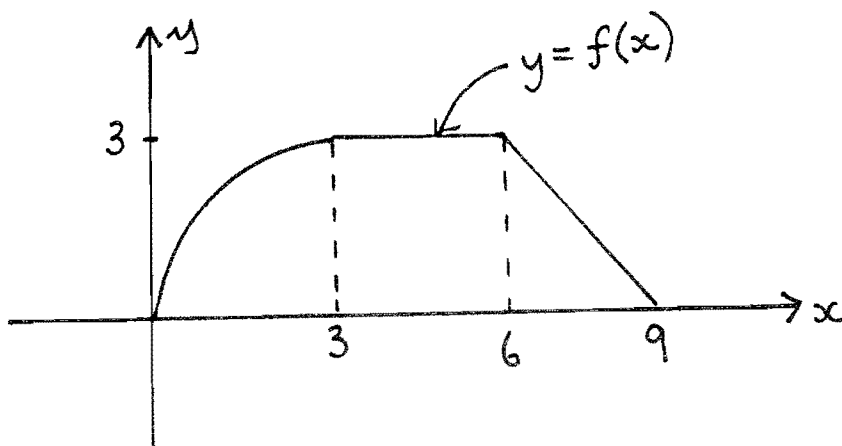
Question 1

(12 marks)

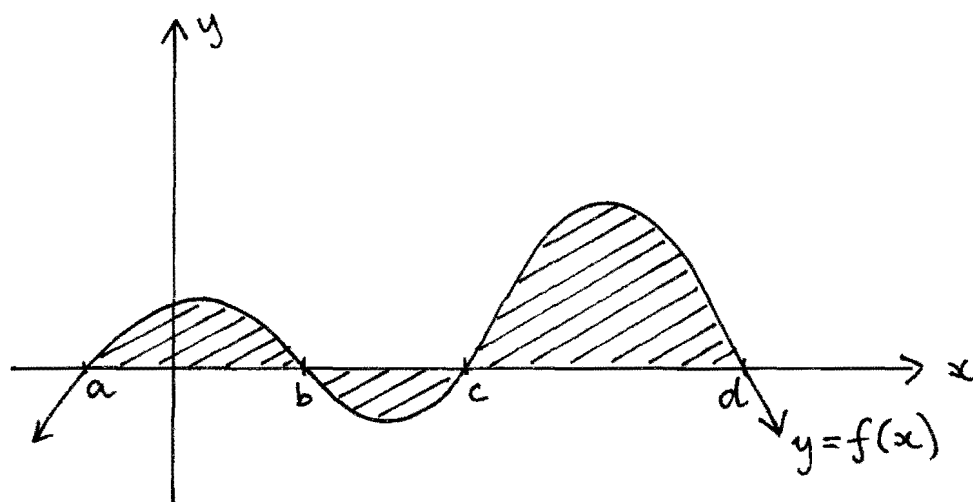
- a) Express 1.45 radians in degrees and minutes (correct to nearest minute) 1
- b) Find the exact value of $\tan \frac{2\pi}{3}$ 1
- c) Solve $\tan x = \sqrt{2} - 1$, for $0^\circ \leq x \leq 360^\circ$ (leaving your answer correct to the nearest minute) 2
- d) i) Express 45° in radians, in terms of π . 1
- ii) Find the area of the shaded section ABCD, below (in terms of π)



- iii) Find the perimeter of the shaded section ABCD, above (in terms of π) 2
- e) Find $\int_0^9 f(x) dx$, given the sketch below (in exact form). 2



f)



To calculate the shaded area above, would the evaluation of $\int_a^d f(x)dx$ give the correct solution? Explain your answer.

1

Question 2 (Start a new page) (12 marks)

- | | | |
|------|---|---|
| a) | Find k , if $\int_0^3 kx^2 dx = 4$ | 2 |
| b) | Evaluate $\log_3 2$, correct to 2 decimal places | 1 |
| c) | Solve $\log_x 27 = \frac{3}{2}$ | 2 |
| d) | Simplify $\log_5 125 - \log_5 \frac{1}{25} - \log_5 \sqrt{5}$ | 2 |
| e) | Differentiate the following: | |
| i) | $y = 3 \ln 5x$ | 1 |
| ii) | $y = \ln(2 - 3x)$ | 1 |
| iii) | $y = e^{2x}$ | 1 |
| iv) | $y = 2\cos 3x$ | 2 |

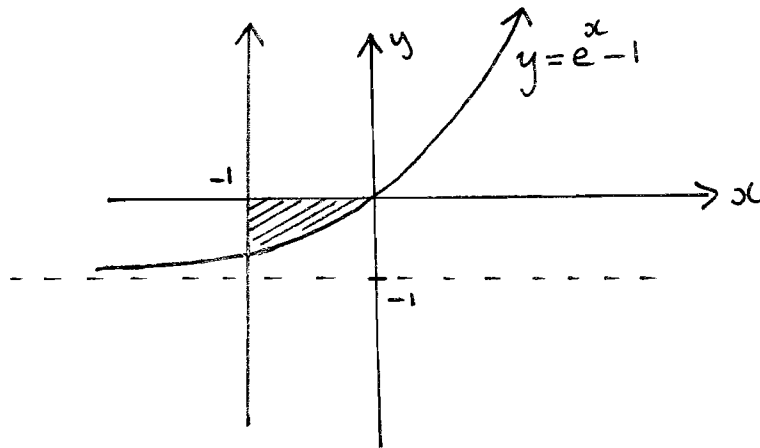
Question 3 (Start a new page) (12 marks)

- | | | |
|----|---|---|
| a) | Find $\frac{d}{dx}(\sqrt{x} \cdot \ln x)$ | 2 |
| b) | Evaluate $\int_1^e \frac{2}{x} dx$ | 2 |
| c) | Find $\int \frac{3-x}{12x-3-2x^2} dx$ | 2 |

- d) Find the equation of the tangent to the curve $y = \tan 2x$, at the point where $x = \frac{\pi}{6}$ 3
- e) The curve $y = \frac{1}{x^2}$ is called a truncus. It is rotated around the y axis from $y = 1$ to $y = 6$. Find the volume of the solid formed (in exact form). 3

Question 4 (Start a new page) (12 marks)

- a) Find $\int e^{7-2x} dx$ 1
- b) Find the area of the shaded section below, that is bounded by the x axis, the curve $y = e^x - 1$, and the line $x = -1$ (in exact form) 3



- c) Differentiate $y = \frac{\cos 2x}{e^x}$ 2
- d) i) Sketch $y = 2\sin 3x$, for $0 \leq x \leq \frac{\pi}{3}$ 2
- ii) Find the area of the region bounded by $y = 2\sin 3x$, and the x axis, in your sketch above. 3
- e) Find $\lim_{x \rightarrow 0} \frac{\sin x}{2x}$ 1

Question 5**(Start a new page)****(12 marks)**

- a) i) Find $\frac{d}{dx}(\sin x - x \cos x)$. 3
- ii) Hence, find $\int_0^{\frac{\pi}{2}} x \cdot \sin x \, dx$. 2
- b) Consider the curve $y = x \ln x$
- i) Find its domain 1
- ii) Find any stationary points on the curve, and determine their nature. 3
- iii) Explain why the curve has **no** points of inflexion. 1
- iv) Sketch the curve, showing any stationary points, and where curve cuts the x and y axes, if it does so. 2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

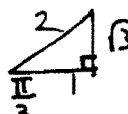
QUESTION 1

$$a) 1.45^\circ = \frac{1.45 \times 180}{\pi}$$

$$= \underline{\underline{83^\circ 5'}}$$

$$b) \tan \frac{2\pi}{3} = \tan \left(\pi - \frac{\pi}{3} \right) \checkmark \begin{array}{c|c} S & A \\ \hline T & C \end{array}$$

$$= -\tan \frac{\pi}{3}$$

$$= \underline{\underline{-\sqrt{3}}}$$


$$c) \tan x = \sqrt{2} - 1$$

$$= .414 \quad \checkmark \begin{array}{c|c} S & A \\ \hline T & C \end{array}$$

$$\therefore x = \underline{\underline{22^\circ 30', 202^\circ 30'}}$$

$$d) i) 45^\circ = \underline{\underline{\frac{\pi}{4}}}$$

$$ii) A = \frac{1}{2} \cdot 15^2 \cdot \frac{\pi}{4} - \frac{1}{2} \cdot 10^2 \cdot \frac{\pi}{4}$$

$$= \frac{225\pi}{8} - \frac{100\pi}{8}$$

$$= \underline{\underline{\frac{125\pi}{8} \text{ cm}^2}}$$

$$iii) P = 10 + 10 \cdot \frac{\pi}{4} + 15 \cdot \frac{\pi}{4}$$

$$= \underline{\underline{\left(\frac{25\pi}{4} + 10 \right) \text{ cm}}}$$

$$e) \int_0^9 f(x) dx = \frac{\pi \cdot 3^2}{4} + 9 + \frac{9}{2}$$

$$= \underline{\underline{\frac{9\pi}{4} + \frac{27}{2}}}$$

f) Incorrect solution - need to take abs. value of area below axis

QUESTION 2

$$a) \int_0^3 kx^2 dx = 4$$

$$\left[\frac{kx^3}{3} \right]_0^3 = 4$$

$$9k - 0 = 4$$

$$9k = 4$$

$$k = \underline{\underline{4/9}}$$

$$b) \log_3 2 = \frac{\log_e 2}{\log_e 3}$$

$$= \underline{\underline{0.63}}$$

$$c) \log_x 27 = \frac{3}{2}$$

$$\therefore x^{3/2} = 27$$

$$x = 27^{2/3}$$

$$x = \underline{\underline{9}}$$

$$d) \log_5 125 - \log_5 \frac{1}{25} - \log_5 \sqrt{5}$$

$$= \log_5 5^3 - \log_5 5^{-2} - \log_5 5^{1/2}$$

$$= 3 \log_5 5 + 2 \log_5 5 - \frac{1}{2} \log_5 5$$

$$= 3 + 2 - \frac{1}{2}$$

$$= \underline{\underline{4\frac{1}{2}}}$$

$$e) i) y = 3 \ln 5x \therefore \frac{dy}{dx} = \underline{\underline{\frac{3}{x}}}$$

$$ii) y = \ln(2-3x) \therefore \frac{dy}{dx} = \underline{\underline{\frac{-3}{2-3x}}}$$

$$iii) y = e^{2x} \therefore \frac{dy}{dx} = \underline{\underline{2e^{2x}}}$$

$$iv) y = 2 \cos 3x \therefore \frac{dy}{dx} = \underline{\underline{-6 \sin 3x}}$$

QUESTION 3

$$\begin{aligned} \text{a) } u &= \sqrt{x} = x^{1/2} & v &= \ln x \\ u' &= \frac{1}{2} x^{-1/2} & v &= \frac{1}{x} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$

$$\therefore \frac{d}{dx} (\sqrt{x} \ln x) = \frac{\ln x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}$$

$$\begin{aligned} \text{b) } e \int \frac{2}{x} dx &= \left[2 \ln x \right]_1^e \\ &= 2 \ln e - 2 \ln 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{c) } \int \frac{3-x}{12x-3-2x^2} dx \\ &= \frac{1}{4} \int \frac{12-4x}{12x-3-2x^2} dx \\ &= \frac{1}{4} \ln (12x-3-2x^2) \end{aligned}$$

$$\begin{aligned} \text{d) } y &= \tan 2x & y &= \tan \frac{\pi}{3} \\ \frac{dy}{dx} &= 2 \sec^2 2x & \therefore y &= \sqrt{3} \end{aligned}$$

$$\text{at } \left(\frac{\pi}{6}, \sqrt{3} \right) \quad m = 2 \sec^2 2 \times \frac{\pi}{6}$$

$$\begin{aligned} m &= 2 \sec^2 \frac{\pi}{3} \\ m &= 8 \end{aligned}$$

$\therefore$ tangent

$$y - \sqrt{3} = 8 \left(x - \frac{\pi}{6} \right)$$

$$\begin{aligned} \text{e) } \text{Graph of } y = \frac{1}{x^2} \text{ from } x=1 \text{ to } x=6. \\ V &= \pi \int_1^6 \frac{1}{y} dy \\ &= \pi \left[\ln y \right]_1^6 \end{aligned}$$

$$\begin{aligned} V &= \pi (\ln 6 - \ln 1) \\ &= \pi \ln 6 \text{ units}^3 \end{aligned}$$

QUESTION 4

$$\text{a) } \int e^{7-2x} dx = \frac{-1}{2} e^{7-2x} + C$$

$$\begin{aligned} \text{b) } A &= \left| \int_{-1}^0 e^x - 1 dx \right| \\ &= \left| \left[e^x - x \right]_{-1}^0 \right| \\ &= \left| (1+0) - (e^{-1} + 1) \right| \\ &= \left| -\frac{1}{e} \right| \\ &= \frac{1}{e} \text{ unit}^2 \end{aligned}$$

$$\begin{aligned} \text{c) } u &= \cos 2x & v &= e^x \\ u' &= -2 \sin 2x & v' &= e^x \\ \frac{dy}{dx} &= \frac{-2e^x \sin 2x - e^x \cos 2x}{e^{2x}} \\ &= \frac{e^x (-2 \sin 2x - \cos 2x)}{e^{2x} e^x} \\ &= \frac{-2 \sin 2x - \cos 2x}{e^x} \end{aligned}$$

$$\text{d) } \text{period } \frac{2\pi}{3} \quad \text{amp} = 2$$

$$\text{i) } \text{Graph of } y = 2 \sin 3x \text{ from } x=0 \text{ to } x=\frac{\pi}{3}.$$

$$\begin{aligned}
 \text{ii) } A &= \int_0^{\pi/3} 2 \sin 3x \, dx \\
 &= \left[-\frac{2}{3} \cos 3x \right]_0^{\pi/3} \\
 &= \left[-\frac{2}{3} \cos \pi - \left(-\frac{2}{3} \cos 0 \right) \right] \\
 &= \left[\frac{2}{3} + \frac{2}{3} \right] \\
 &= \underline{\underline{\frac{4}{3} \text{ units}^2}}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \lim_{x \rightarrow 0} \frac{\sin x}{2x} \\
 \lim_{x \rightarrow 0} \frac{1}{2} \cdot \frac{\sin x}{x} \cdot \frac{1}{2} \\
 = \underline{\underline{\frac{1}{2}}}
 \end{aligned}$$

QUESTION 5

$$\begin{aligned}
 \text{a) i) } u &= -x & v &= \cos x \\
 u' &= -1 & v' &= -\sin x
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{d}{dx} (\sin x - x \cos x) \\
 &= \cos x - \cos x + x \sin x \\
 &= \underline{\underline{x \sin x}}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \int_0^{\pi/2} x \sin x \, dx \\
 &= \left[\sin x - x \cos x \right]_0^{\pi/2}
 \end{aligned}$$

$$\begin{aligned}
 &= \left(\sin \frac{\pi}{2} - \frac{\pi}{2} \cos \frac{\pi}{2} \right) - (0) \\
 &= \underline{\underline{1}}
 \end{aligned}$$

$$\text{b) } y = x \ln x$$

$$\text{i) } \underline{x > 0}$$

$$\begin{aligned}
 \text{ii) } u &= x & v &= \ln x \\
 u' &= 1 & v' &= \frac{1}{x}
 \end{aligned}$$

$$\frac{dy}{dx} = \ln x + 1$$

$$\frac{d^2y}{dx^2} = \frac{1}{x}$$

$$\begin{aligned}
 \text{st pts } y' &= 0 & \ln x + 1 &= 0 \\
 \log_e x &= -1
 \end{aligned}$$

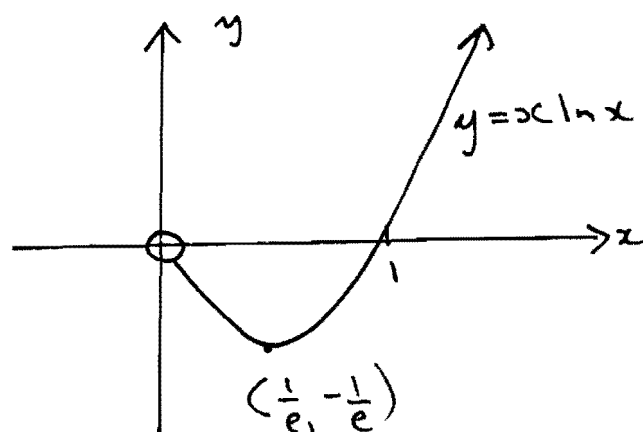
$$\therefore x = e^{-1}$$

$$\text{at } \left(\frac{1}{e}, -\frac{1}{e} \right) y'' > 0 \therefore \text{min}$$

$$\begin{aligned}
 \text{if } x &= \frac{1}{e} & y &= \frac{1}{e} \ln e^{-1} \\
 & & y &= -\frac{1}{e}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) pt inf } y'' &= 0 \\
 \text{since } \frac{1}{x} &\neq 0 \therefore \text{no pt inf.}
 \end{aligned}$$

iv)



$$y = 0$$

$$\begin{aligned}
 x \cdot \ln x &= 0 \\
 x \neq 0 & \ln x = 0 \\
 \log_e x &= 0
 \end{aligned}$$

$$e^0 = x$$

$$\therefore x = 1$$

$$\underline{\underline{\text{cut } x \text{ axis at } x=1}}$$